

Innovation Declaration

The final submitted strategy is `phase2_g5_05_expanding`: a volatility-scaled close-to-open target with expanding-window transparent feature-weight estimation.

What Changed Versus Baseline

The original baseline predicted raw next overnight returns using the provided point-in-time feature set. The final strategy keeps that feature set unchanged but trains on a risk-adjusted label:

`overnight_next / (vol120 / sqrt(252))`

where `vol120` is trailing 20-day close-to-close volatility shifted by one trading day. The promoted Phase 2 improvement changes the training window from the previous 4-year rolling champion to an expanding window. For each scored year, the model uses only earlier years and never includes the year being scored.

Why This Is An Innovation

The change is transparent and economically motivated. Volatility scaling reduces the influence of noisy high-volatility overnight labels. Expanding-window estimation uses more accumulated overnight evidence as time progresses, which stabilises the simple feature-weight learner without adding opaque modelling complexity.

What Did Not Change

The promotion did not use external data, intraday data, 2025+ data, new feature families, lower transaction costs, lower borrow costs, a relaxed participation cap, or different execution timing. Commission, auction slippage, Tier A/B/C borrow costs, the 5% ADV20 cap, and close-to-open execution all remain fixed.

Evidence

At 250M, validation 2019-2022 net Sharpe is 2.279, internal holdout 2023-2024 net Sharpe is 0.620, and full 2010-2024 net Sharpe is 1.445. The holdout result is lower than validation, so the result is presented as a controlled improvement rather than an overfit-free guarantee.